



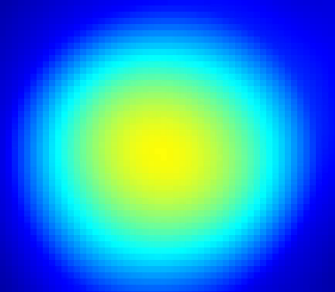
Internship at Institut ∂ 'Alembert

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Simulation of multiphase flows

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Abstract

In this paper, we will use the CFD software *Basilisk* to simulate multiphase flows and investigate the thermal and dynamic interactions between a droplet and a surrounding fluid. We compare three formulations of the heat transfer problem, study the influence of the Reynolds and Péclet numbers, and evaluate their convergence for a static droplet when refining the mesh.

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1 Introduction

1.1 Useful constants for our problem

There are many parameters to evaluate if we want to produce a good enough simulation : the masses of the liquid bodies, their volume, temperature, viscosity, density, etc.

In order to simplify the simulations as much as we can, we have to reduce the number of constants used. To do so, we will redefine them from three characteristics of the problem : the speed of the droplet u_∞ , its radius R_0 and its density ρ_d . With that, we can define some characteristic quantities :

- The characteristic length L_c , speed u_c and density ρ_c are, respectively $10R_0$, u_∞ and ρ_d .
- We also need to define the volumetric heat capacity $\rho_c C_{P_c}$, which is characteristic of the droplet and is therefore equal to $\rho_d C_{P_d}$.
- From there, we can define the characteristic time t_c of our simulation, which represents the time needed for the droplet to move significantly. This time is defined as:

$$t_c = \frac{L_c}{u_c} = \frac{10R_0}{u_\infty}$$

- We can also find the main characteristics of the droplet with these three parameters : its mass is given by $m_c = \rho_c L_c^3 = 1000\rho_d R_0^3$, its viscosity by $\mu_c = \rho_c u_c L_c = 10\rho_d u_\infty R_0$, its acceleration by $a_c = \frac{u_c}{t_c} = \frac{u_\infty^2}{10R_0}$.
- The movement of the droplet in the fluid is slowed down by a characteristic drag force F_c , whose expression is given by:

$$F_c = \rho_c u_c^2 L_c^2 = 100\rho_d u_\infty^2 R_0^2$$

- This drag force allows us to express the pressure P_c exerted on the droplet. Indeed, the pressure being a force per unit area, its expression is given by:

$$P_c = \frac{F_c}{L_c^2} = \rho_d u_\infty^2$$

Finally, we introduce 3 numbers that will have an importance in our simulations. Firstly, the Reynolds number. It measures the ratio between the inertial forces and the viscous forces. It is defined by

$$Re = \frac{\rho_c L_c u_\infty}{\mu}$$

Secondly, the Weber number. This number is a measure of the relative importance of the fluid's inertia compared to its surface tension. It relies on Laplace pressure P_L , whose expression is $P_L = \frac{\gamma}{L_c} = \frac{\gamma}{10R_0}$, where γ is the surface tension between the two liquids. The Weber number is thus defined as:

$$We = \frac{P_c}{P_L} = \frac{10\rho_d R_0 u_\infty^2}{\gamma}$$

Thirdly, the Péclet number. It measures the ratio between the advective transport rate and the diffusive transport rate. Its expression is thus given by

$$Pe = \frac{\rho_d C_{P_d} u_\infty L_c}{\kappa}$$

1.2 From the physical world to our simulation

In the first part, we defined the constants used in our simulation. But these constants have a dimension whereas a computer program only takes dimensionless parameters. To solve this issue, the method used here consists in fixing the values of the quantities of one fluid, and determining those of the second fluid relatively to the first one. For simplicity, we will set the values of u_∞^* and ρ_d^* to 1; and the value of R_0^{*1} to 0.1. We name the droplet “fluid 1” and the surrounding fluid “fluid 2”: therefore, if a constant is subscribed with the number 1, this constant is relative to the droplet, and it is similar for the surrounding fluid. We then set a dimensionless drag force F^* the droplet is subject to, as well as its dimensionless viscosity μ_1^* . These parameters are defined by:

$$F^* = \frac{F_c}{\rho_d u_\infty^2 R_0^2} = C_0$$

$$\mu_1^* = \frac{\mu_1}{\rho_d u_\infty R_0} = \frac{1}{Re}$$

where C_0 is a constant ($C_0 = 0$ for the rest of this paper), and Re is the Reynolds number.

Using the same method, we define as well a dimensionless surface tension γ^* and a dimensionless thermal conductivity κ^* using Weber and Péclet's numbers:

$$\gamma^* = \frac{\gamma}{\rho_c u_c^2 L_c} = \frac{1}{We}$$

$$\kappa_1^* = \frac{\kappa_1}{\rho_1 C_{P_1} u_\infty L_c} = \frac{1}{Pe}$$

We also define several ratios: r_ρ , r_μ , $r_{\rho C_P}$ and r_κ . r_ρ is the ratio between the density of the droplet and the density of the surrounding fluid, r_μ is the ratio between the viscosity of the droplet and the viscosity of the surrounding fluid, $r_{\rho C_P}$ is the ratio of volumetric heat capacity of fluid 1 and fluid 2, and r_κ is the ratio of the thermal conductivity of both fluids. These ratios allow us to easily determine the dimensionless density and viscosity of the 2nd fluid, by the formulas:

$$\rho_2^* = \frac{\rho_1^*}{r_\rho}$$

$$\rho_2 C_{P_2}^* = \frac{\rho_1 C_{P_1}^*}{r_{\rho C_P}}$$

$$\mu_2^* = \frac{\mu_1^*}{r_\mu}$$

$$\kappa_2^* = \frac{\kappa_1^*}{r_\kappa}$$

¹The * indicates that the values are dimensionless.

And because $\rho_1^* = \frac{\rho_1}{\rho_c} = 1$, and $\rho_1 C_{P_1}^* = \frac{\rho_1 C_{P_1}}{\rho_c C_{P_c}} = 1$, the first two expressions can be simplified in:

$$\rho_2^* = \frac{1}{r_\rho}$$

$$\rho_1 C_{P_1}^* = \frac{1}{r_{\rho C_P}}$$

2 Numerical models

2.1 Reference problem

We want to obtain the numerical solution of the heat equation with discontinuous coefficients:

$$\rho_i c_{p,i} \left(\frac{\partial T_i}{\partial t} + \mathbf{u} \cdot \nabla T_i \right) = \nabla \cdot (\kappa_i \nabla T_i) \quad (1)$$

with $i=1,2$. The general jump conditions if we take the temperature as primitive variable are:

$$[T] = 0, \quad [\kappa \nabla T \cdot \mathbf{n}_I] = 0, \quad \mathbf{x} = \mathbf{x}_I \quad (2)$$

where $\mathbf{n}_I$ is the normal to the interface located at $\mathbf{x}_I$ and defined positive when pointing towards fluid 2 and the jump is defined as $[T] = T_2(\mathbf{x}_I) - T_1(\mathbf{x}_I)$. Typically the equation above is discretized as follows:

$$\frac{T_i^{(*)} - T_i^{(t)}}{\Delta t} + \nabla \cdot (\mathbf{u} T_i) = 0 \quad (3)$$

$$\rho_i c_{p,i} \frac{T_i^{(t+\Delta t)} - T_i^{(*)}}{\Delta t} = \nabla \cdot (\kappa_i \nabla T_i^{(t+\Delta t)}) \quad (4)$$

This is called time-splitting technique.

2.2 Regularized methods

2.2.1 Standard implementation

The naive version of the regularization method is:

$$\frac{\tilde{T}^{(*)} - \tilde{T}^{(t)}}{\Delta t} + \nabla \cdot (\mathbf{u} \tilde{T}) = 0 \quad (5)$$

$$\frac{\tilde{T}^{(t+\Delta t)} - \tilde{T}^{(*)}}{\Delta t} = \nabla \cdot (\tilde{\kappa} \nabla \tilde{T}^{(t+\Delta t)}) \quad (6)$$

We will use this solution as reference. VOF is used to represent the interface position.

2.2.2 Sharp interface reconstruction method

We propose to use:

$$\frac{h_i^{(*)} - h_i^{(t)}}{\Delta t} + \nabla \cdot (\mathbf{u} h_i) = 0 \quad (7)$$

$$\frac{\tilde{T}^{(t+\Delta t)} - T_i^{(*)}}{\Delta t} = \nabla \cdot ((\tilde{\kappa} \nabla \tilde{T})^{(t+\Delta t)}) \quad (8)$$

where $h_i = \rho_i c_{p,i} T_i$ is the enthalpy.

The coefficient $\tilde{\kappa}$ varies $[\kappa] = \kappa_2 - \kappa_1$ over a characteristic small distance h_{reg} which spatial structure depends on a regularization function $f(\xi)$ bounded between 0 and 1. In this work we will use:

$$\tilde{\kappa} = \kappa_1 f + \kappa_2 (1 - f), \quad \frac{1}{\tilde{\kappa}} = \frac{f}{\kappa_1} + \frac{1-f}{\kappa_2}$$

The regularization function depends on the signed distance function ξ given by the Eikonal equation $|\nabla \xi| = 1$. The function $f(\xi)$ only varies in a compact and narrow region with a thickness $|\xi| \leq h_{\text{reg}}/2$ around the exact position of the sharp interface. Without lack of generality, we define:

$$f = \begin{cases} 1, & \xi \leq \xi_1 \\ 0.5 - \frac{\xi}{h_{\text{reg}}}, & \xi_1 \leq \xi \leq \xi_2 \\ 0, & \xi > \xi_2 \end{cases} \quad (9)$$

with $\xi_1 = -h_{\text{reg}}/2$ and $\xi_2 = h_{\text{reg}}/2$.

3 First results

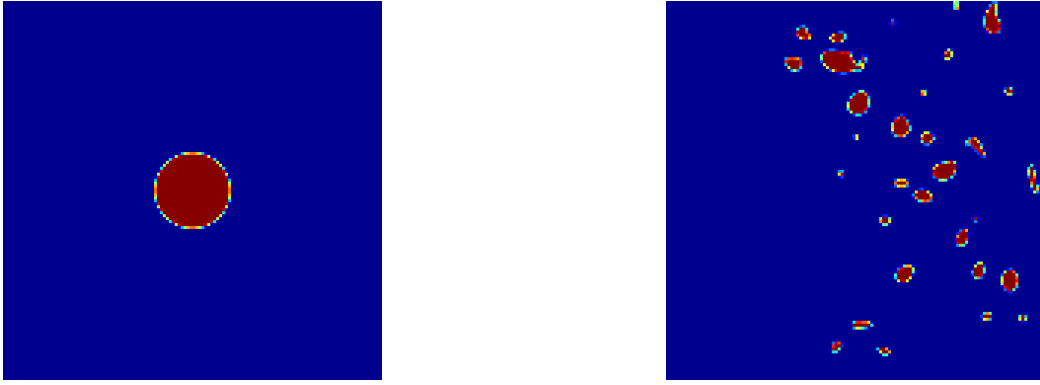
To model the interactions between the droplet and the surrounding fluid, we will use and compare the three regularized methods developed in the section 2.1 and 2.2. In order for our simulation to be able to work, we have to define Reynolds's, Péclet's and Weber's numbers, as well as our several ratios: r_ρ , $r_{\rho C_P}$, r_μ and r_κ .

3.1 Result for a broad example

For this first example, we're going to simulate the interaction of a bubble of air inside water. According to experimental values, we set our different constants to:

$$r_\rho = \frac{1.117}{997.0}, \quad r_{\rho C_P} = \frac{1.123}{4172.4}, \quad r_\mu = \frac{0.0000185}{0.001}, \quad r_\kappa = \frac{0.026}{0.6}$$

Also, to keep our problem relatively simple, we set ρ_1 and $\rho_{C_{P_1}}$ to 1, T_1 to 1 and T_2 to 0. We also set Reynolds' number to 100, Péclet's number to 10 and Weber's number to 5. After letting the code run for a certain amount of time, we remark an important evolution of the bubble:

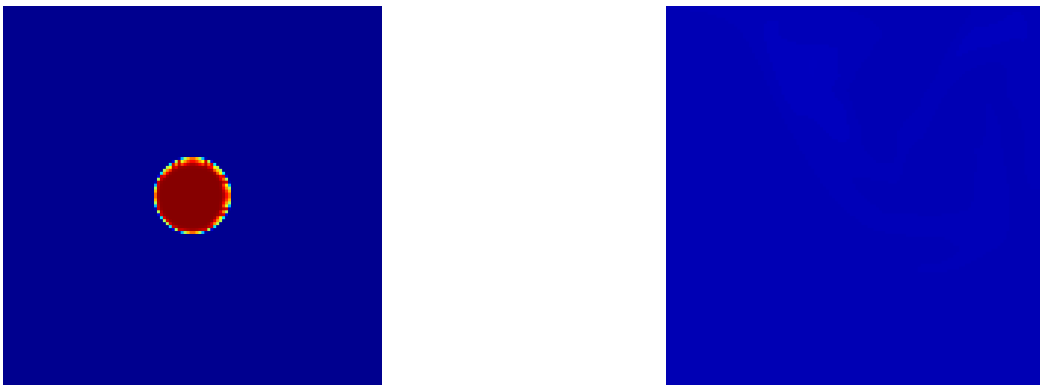


(a) Air bubble at the beginning of the simulation.

(b) Air bubble towards the end of the simulation.

Figure 1: Evolution of the air bubble's position in time.

We observe that over time, the air bubble deforms significantly and eventually shatters into several smaller bubbles of high temperature. This fragmentation process is primarily caused by the high density contrast between the air and the surrounding water, as well as interfacial shear stress. As the bubble interacts with the fluid, the hydrodynamic inertial forces are stronger than the cohesive effect of surface tension. This effect is stronger as we choose a bigger number of Weber. Because of that, interfacial instabilities develop around the bubble's boundary, leading to its collapse. We then study the evolution of the temperature inside the system:



(a) Temperature in the system at the beginning of the simulation.

(b) Temperature in the system towards the end of the simulation (with increased contrast).

Figure 2: Evolution of the air bubble's temperature in time.

We clearly notice that the thermal energy, initially concentrated within the air bubble, is advected and diffused into the surrounding cold water as the bubble shatters, because of interfacial instabilities. This fragmentation creates multiple localized regions of higher temperature that will quickly equilibrate with the colder fluid.

As a matter of fact, we can determine this equilibrium temperature T_{eq} , which satisfies the relation:

$$T_{eq} = \frac{V_1 \rho_1 C_{P_1} T_1 + V_2 \rho_2 C_{P_2} T_2}{V_1 \rho_1 C_{P_1} + V_2 \rho_2 C_{P_2}} \quad (10)$$

By replacing these values with those we set earlier, we get that:

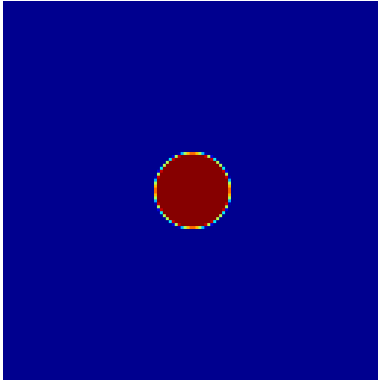
$$T_{eq} = \frac{1}{1 + \frac{4172.4}{1.123} \left(\frac{100}{\pi} - 1 \right)} \simeq 8.7 \times 10^{-6}$$

3.2 Result for another example

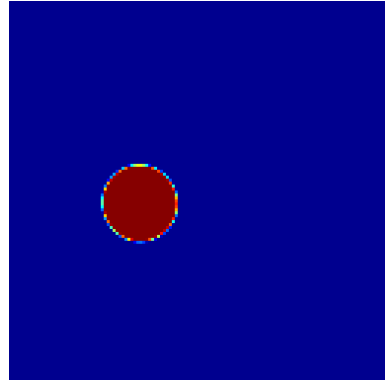
For this second simulation, we will this time model the interaction between an oil droplet and the water surrounding it. Once again, experimental works are such that we define our constants to:

$$r_\rho = \frac{918.0}{997.0}, \quad r_{\rho C_P} = \frac{1836.0}{4172.4}, \quad r_\mu = \frac{0.084}{0.001}, \quad r_\kappa = \frac{0.17}{0.6}$$

We then set ρ_1 and $\rho_{C_{P_1}}$ to 1, T_1 to 1 and T_2 to 0; Reynolds' number to 100, Péclet's number to 10 and Weber's number to 5, just like the previous simulation. After letting our code run, we obtain the following results:



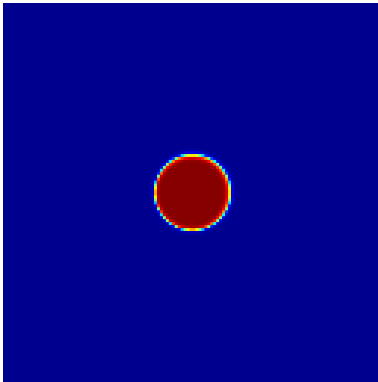
(a) Oil bubble at the beginning of the simulation.



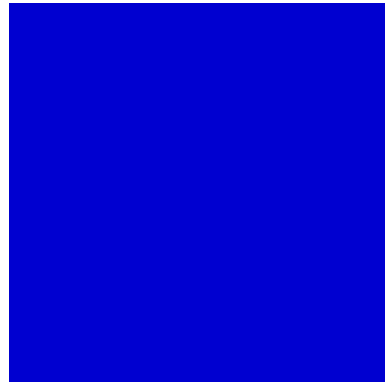
(b) Oil bubble towards the end of the simulation.

Figure 3: Evolution of the oil bubble's position in time.

This time, the bubble did not shatter. That is because oil possesses a much higher density and viscosity compared to air. As such, the interfacial instabilities are weakened and the bubble doesn't breakup. Once again, we study the evolution of the temperature inside the system:



(a) Temperature in the system at the beginning of the simulation.



(b) Temperature in the system towards the end of the simulation (with increased contrast).

Figure 4: Evolution of the oil bubble's temperature in time.

Like for the air bubble, the temperature of the oil droplet diffuses progressively into the surrounding water over time, eventually reaching a uniform thermal equilibrium across the entire system. The equilibrium temperature is in this case approximately equal to 1.4×10^{-2} .

Because of the stability of the oil droplet in the water compared to the air bubble, we will from now on use this case to do our future experiments.

4 Convergence of the models

Our simulations display the solutions found by different models to the heat equation. Our goal is now to find how fast the solutions converge to their “optimal” solutions. For each situation we will thus give the results obtained when calculating the enthalpy (method ①), when considering the fluid system is one fluid with different temperatures (method ②), and when considering the temperature of the two liquid bodies (method ③).

4.1 Influence of Reynolds’ number over the solution

We first take a look at how the temperature evolves when we modify the Reynolds’ number. To achieve this, we will see how the solution differs for $Re = 100$, $Re = 1000$ and $Re = 10000$.

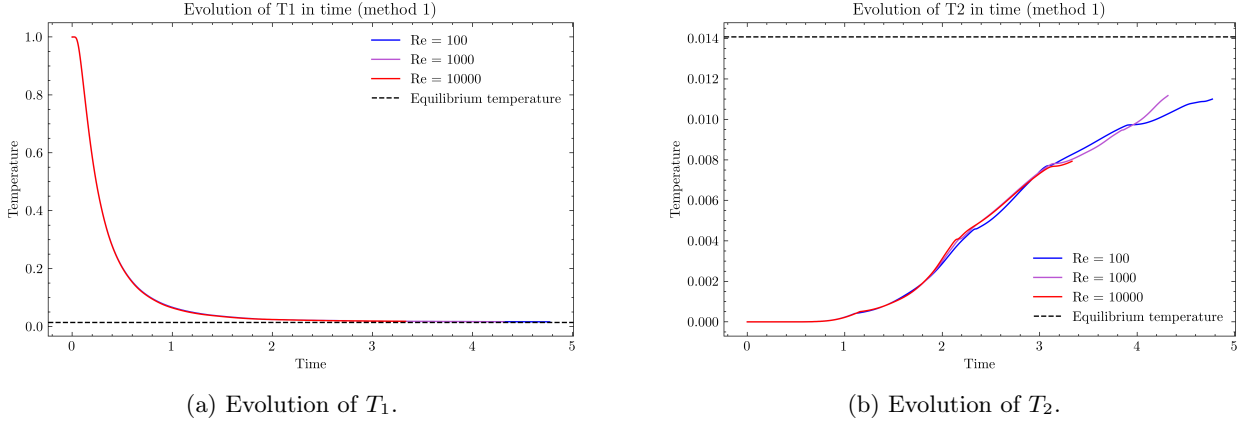


Figure 5: Evolution of the temperatures depending on Reynolds’ number when using method ①.

First, we look at its evolution for method ①. For this method, we observe that the higher is Re , the quicker the solution converges to the equilibrium temperature.

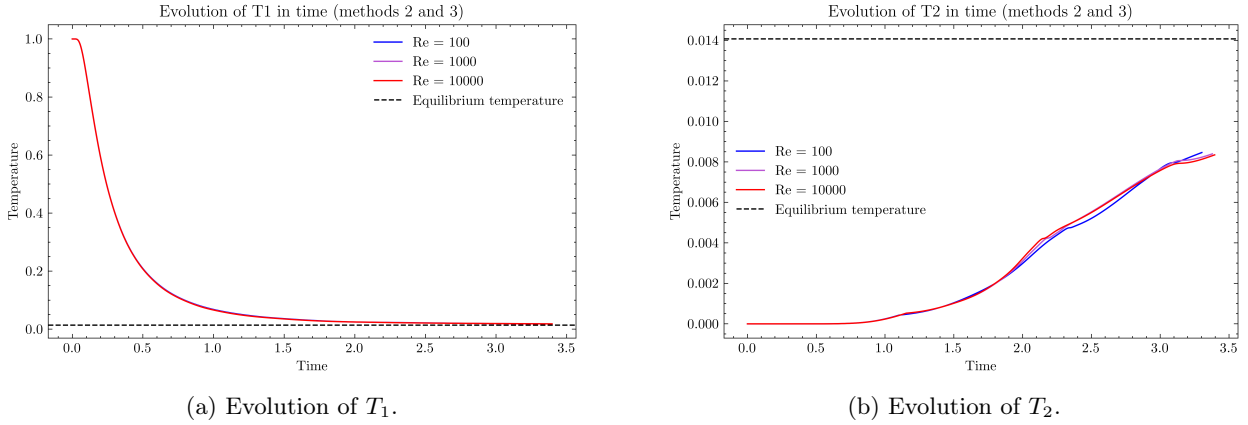
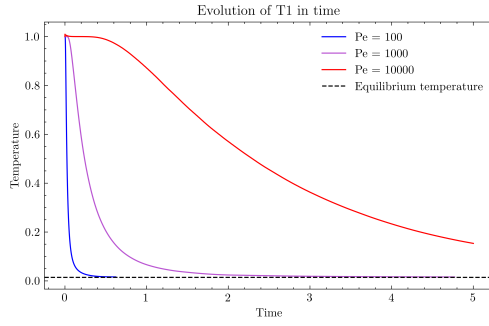


Figure 6: Evolution of the temperatures depending on Reynolds’ number when using method ② and ③.

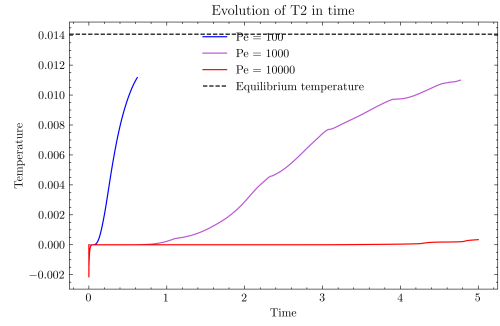
Yet by looking to its evolution for methods ② and ③, we notice that Reynolds’ number has no notable influence over the solutions. A hypothesis we can formulate to explain this phenomenon, resides in the difference in the temperature’s construction. While method ① reconstructs its temperature from enthalpy, obtained from the VOF reconstruction which is highly sensitive to advection; methods ② and ③ solve the temperature field directly, supposedly less subject of minor variations in the velocity field induced by Re . Furthermore, we must recall that the Péclet number is the key factor for the heat equation, while the Reynolds number mostly affects the velocity field. This is why we will now look at the influence of the Péclet number over temperature.

4.2 Influence of Péclet’s number over the solution

After studying the influence of Reynolds’ number on the temperature, we now study the influence of Péclet’s number over it. To keep it relatively simple, we will use the following of Pe : $Pe = 10$, $Pe = 100$ and $Pe = 1000$. Unlike Reynolds’ number, Péclet’s number has an influence no matter the method used, and this one is the same for all methods.



(a) Evolution of T_1 .



(b) Evolution of T_2 .

Figure 7: Evolution of the temperatures depending on Péclet's number.

The plots suggest that increasing the value of Pe strongly increases the time necessary for the temperature to converge to its equilibrium temperature. This phenomenon is explainable as the Péclet number represents the ratio of advective heat transport to diffusive heat transport. A higher Pe indicates that heat diffusion is very weak compared to the flow's advection, thus drastically slowing down the conductive heat transfer between the droplet and the surrounding fluid, and increasing the time required to reach thermal equilibrium.

4.3 Convergence of the temperature for a static droplet

In Basilisk, the spatial domain is discretized using a grid. The resolution of this mesh is controlled by the `LEVEL` parameter: in our case, the spatial domain is fixed to $[-0.5, 0.5] \times [-0.5, 0.5]$ and is discretized into a grid of $2^{LEVEL} \times 2^{LEVEL}$. The size of a cell dx is thus given by $dx = \frac{|[-0.5, 0.5]|}{2^{LEVEL}} = 2^{-LEVEL}$. Our goal is to understand how evolves the temperature in the system when we raise `LEVEL`. To do so, we model a static droplet. Because of this, the droplet is speedless and therefore, $Re = Pe = We = 0$. Thus, we cannot define μ and ρ , and we set κ_1 and ρC_P to 1 as they are still needed for calculations. We toss the values of r_μ and r_ρ as a consequence of the previous sentence, and we keep r_κ and $r_{\rho C_P}$ the same.

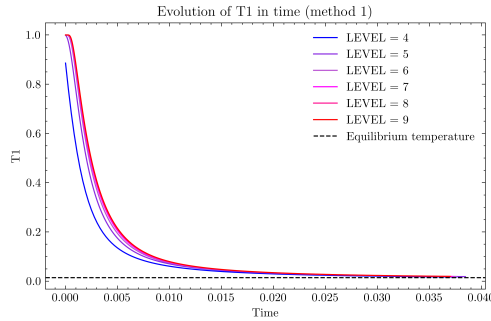
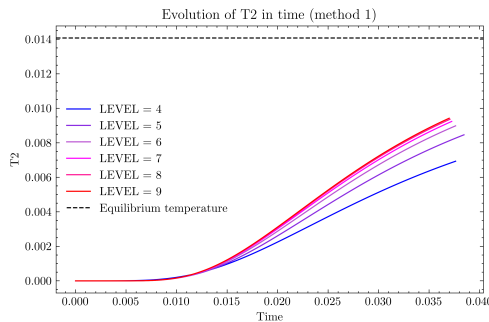
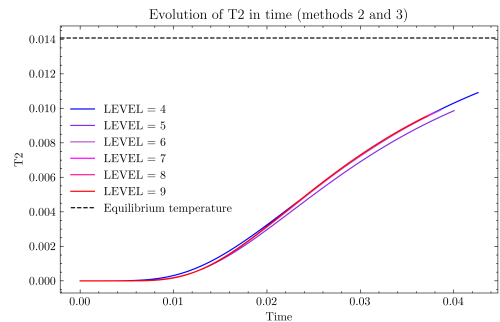


Figure 8: Evolution of T_1 according to the `LEVEL`.

After letting the code run, it seems that by increasing the value of `LEVEL`, T_1 converges towards an “optimal” solution, the curves for `LEVEL` = 8 and `LEVEL` = 9 being almost identical. Because $Pe = 0$, the diffusive transport is infinitely more prominent than the advective transport. The time scale thus changed a lot, but isn't “shorter” nevertheless.



(a) Using method ①.



(b) Using methods ② and ③.

Figure 9: Evolution of T_2 according to the `LEVEL`.

Yet it appears that T_2 is not converging in the same way relatively to the method used: for method ①, T_2 increases gradually when increasing the value of LEVEL, but for methods ② and ③, it stagnates around the actual “optimal” solution. To verify which of these three methods give better results, we evaluate the error generated when modifying the grid size:

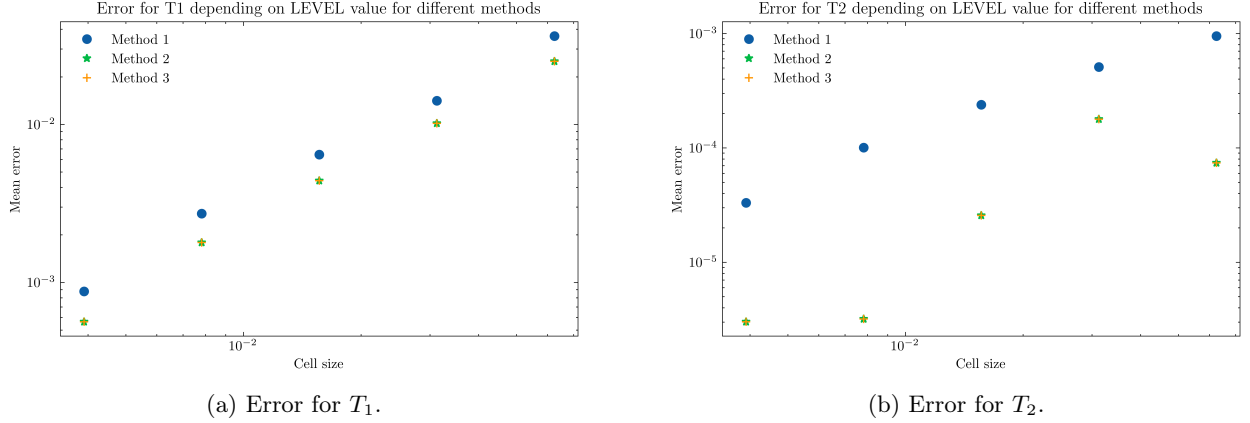


Figure 10: Evolution of the error depending on the method used.

Looking at the graphs, we see that no matter the temperature studied, methods ② and ③ appear to give better results. As a matter of fact, these two methods appear to be rigorously equivalent, as they give the exact same results every single time. And overall, increasing the size of the grid, thus reducing the size of a cell, allows to reduce the error.

While these results might seem off at first glance, as such a difference of convergence between these three methods is quite unexpected, the overall temperature still successfully converges to the same “optimal” solution no matter the method used, as shown in figure 11:

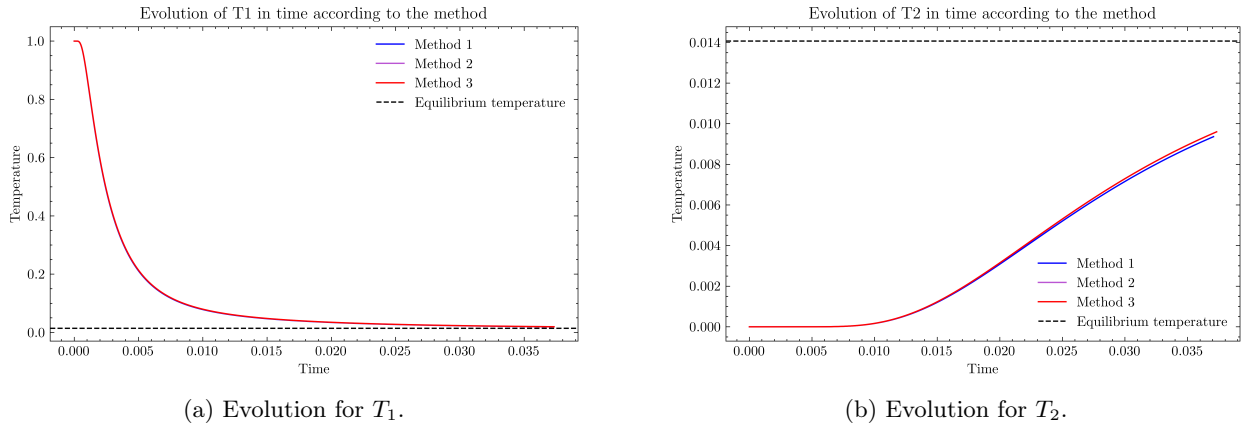


Figure 11: Evolution of the temperatures depending on the method used.

5 Comparison with a simplified calculation method

The method presented here is a particular case of the sharp interface reconstruction method introduced in section 2.2.2, this time based off a distance. Let’s consider a cell of coordinates (x, y) . Its distance to the droplet is given by $d = \sqrt{x^2 + y^2} - R_0$. By considering Δ_1 a fixed reference length, independent of the grid resolution, we define conditions to determine the smoothed volume fraction sf of the droplet in a cell:

$$sf = \begin{cases} 0, & d \geq \frac{\Delta_1}{2} \\ 0.5 - \frac{d}{\Delta_1}, & -\frac{\Delta_1}{2} < d < \frac{\Delta_1}{2} \\ 1, & d \leq -\frac{\Delta_1}{2} \end{cases} \quad (11)$$

We will use this method to calculate the temperature evolution of a static bubble to then determine the error produced by each method. Because the droplet has no velocity, the distance will remain constant through the whole simulation and doesn’t need to be calculated again, saving a significant amount of time. Thus, our reference model is the solution

obtained using the VOF approach that we got in the previous section, for a LEVEL of 9.

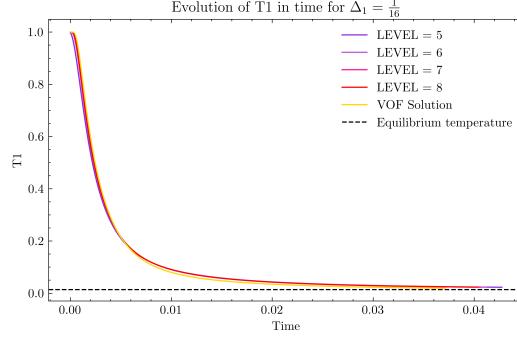
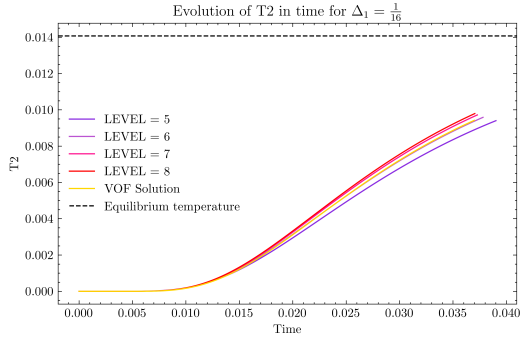


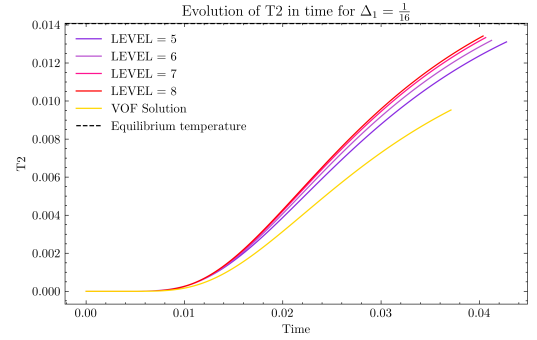
Figure 12: Evolution of T_1 according to the LEVEL for $\Delta_1 = \frac{1}{16}$.

Here, Δ_1 is fixed to a reference value of $\Delta_1 = \frac{1}{16}$, independently of the LEVEL used for the simulation. Fixing it allows us to control how smooth the distance evolves, which is essential to isolate the discretization error from the effect of the smoothing itself. For our study, we thus only consider LEVEL values such that $2^{-\text{LEVEL}} < \Delta_1$, making sure the interface remains resolved by several grid cells.

The solutions are more or less converging to the reference solution for T_1 , though a few differences can be spotted: our reference solution lasts shorter in time, and gets closer to the equilibrium temperature quicker.



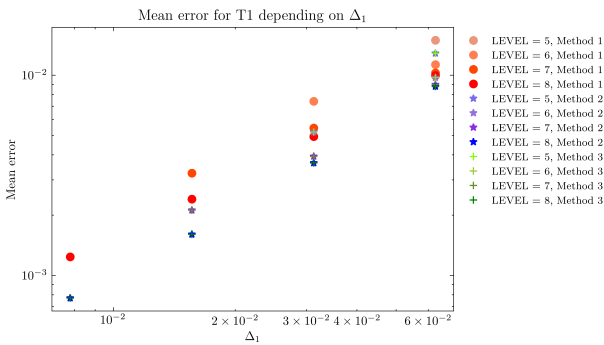
(a) Using method ①.



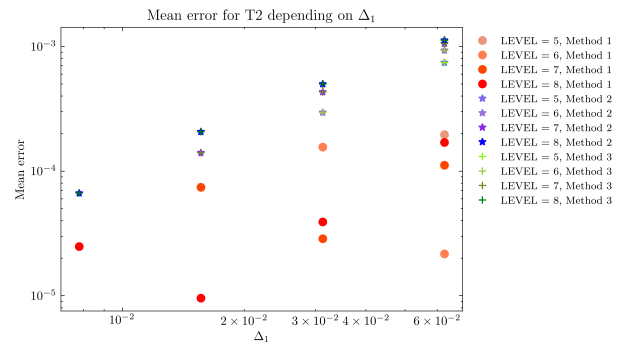
(b) Using methods ② and ③.

Figure 13: Evolution of T_2 depending on the method used.

But as we can see on figure 13, both solutions don't really seem to converge towards the reference solution: for method ①, the VOF solution is in between two other curves, and for methods ② and ③, the VOF solution is lower than any other solutions, and these solutions are getting further and further the more we increase the value of LEVEL.



(a) Error for T_1



(b) Error for T_2

Figure 14: Mean error for temperatures depending on Δ_1

Looking at figure 14, we can notice that once again, methods ② and ③ produce less errors for T_1 , but surprisingly this time, method ① is producing less errors for T_2 . Also, the graph for T_2 is weird looking for method ① because the reference solution is in between two other curves, such that one solution is really close to it while the others span across it. Finally, it seems that increasing the value of LEVEL reduces the error for T_1 as expected, yet it increases the error for T_2 , which is quite odd. It is possible that the temperature is converging towards another “optimal” solution and thus produces more errors as we increase the value of LEVEL, now, but we cannot confirm it yet and it would require further investigation.

6 Conclusion

To conclude, this work successfully simulated multiphase flows and investigated the thermal interactions between a droplet and a surrounding fluid using the software *Basilisk*. By studying the influence of Reynolds’ and Péclet’s numbers, we confirmed that advection plays a critical role in the thermal homogenization process, and significantly dictates the time required to reach thermal equilibrium.

Furthermore, our simulations showed that tracking the temperatures of both fluids (method ③), and tracking the temperature of a single fluid with different temperatures (method ②), is rigorously the same. These two methods also appear to work better than measuring the enthalpy to deduce the temperature (method ①) as overall, they produce less errors.

With these elements, the next goal would be to determine the value of the error as the LEVEL tends to infinity, for all three methods.

References

- [1] John D. Garrett. “garrettj403/SciencePlots”. (For better looking plots). Version 1.0.9. In: (Sept. 2021). DOI: [10.5281/zenodo.4106649](https://doi.org/10.5281/zenodo.4106649). URL: <http://doi.org/10.5281/zenodo.4106649>.